

MAT 1272 Statistics

Group Activities: Normal Distributions and Sampling Distributions

Group Activity 1: Normal Distributions and the TI-84

Instructions: Show all work and calculator commands used. Be prepared to explain your reasoning to the class. Write all the group members (please include last names)

Problem 1

The heights of adult women are approximately normally distributed with a mean height of 64 inches and a standard deviation of 3 inches.

A clothing company is designing a special line of jackets for taller women. They would like to estimate the proportion of women who are taller than 68 inches.

- (a) Define the random variable X .
- (b) Write the probability statement that represents the situation.
- (c) Use the TI-84 calculator to find the probability.
- (d) Interpret your answer in a complete sentence.

Problem 2

Exam scores in a statistics class are approximately normally distributed with $\mu = 75$ and $\sigma = 8$. A student receives a score of 88.

- (a) Compute the student's z-score.
- (b) Is this score above or below the mean?
- (c) Approximately how many standard deviations from the mean is this score?

Problem 3

SAT scores are approximately normally distributed with $\mu = 500$ and $\sigma = 100$. Find the score that separates the lowest 10% of all students.

- (a) Write the percentile being requested.
- (b) Identify the TI-84 command needed.
- (c) Use the TI-84 calculator to find the score.
- (d) Interpret your answer.

Group Activity 2: Sampling Distributions and the Central Limit Theorem

Instructions: Work in groups of 2–3 students. Show all work and calculator commands used. Discuss your reasoning with your group members before writing your final answers.

Problem 1

A population has mean $\mu = 80$ and standard deviation $\sigma = 15$.

A random sample of size

$$n = 25$$

is selected.

Find:

- (a) The mean of the sampling distribution, $\mu_{\bar{x}}$.
- (b) The standard deviation of the sampling distribution, $\sigma_{\bar{x}}$.
- (c) Explain what $\sigma_{\bar{x}}$ represents.

Problem 2

Using the information from Problem 1:

$$\mu = 80, \quad \sigma = 15, \quad n = 25,$$

find the probability that the sample mean exceeds 84.

- (a) Write the probability statement.
- (b) Compute the standard error.
- (c) Use the TI-84 calculator to find the probability.
- (d) Interpret your answer.

Be prepared to share your group's answers with the class.